A Guide to Elliptic Curve Invariants: Understanding the Four Tables

Abstract

We explain four tables of arithmetic and modular data for elliptic curves over $\mathbb{Q}$ with conductor $N \leq 54$, aimed at mathematicians with general background but limited number theory experience. We develop the theory from basics: elliptic curves as algebraic groups, the Mordell-Weil theorem, modular forms and the Modularity Theorem, Galois representations, and why representation-theoretic data matters for understanding L-functions and arithmetic. All technical terms are defined before use.

Contents

1 Introduction: What Are We Looking At? 3

2 Background: Elliptic Curves and Their Groups 3
2.1 Definition and Basic Structure . . . 3
2.2 The j-Invariant . . . 4
2.3 The Conductor . . . 5
2.4 The Cremona Label . . . 6

3 Table 1: Basic Arithmetic Invariants 6
3.1 Column A: Curve Label . . . 6
3.2 Column B: Weight . . . 6
3.2.1 Newforms, Oldforms, and Levels . . . 6
3.3 Column C: Torsion Order . . . 8
3.4 Column D: Torsion Structure . . . 8
3.5 Column E: Rank . . . 9

4 Table 2: Mordell-Weil Generators and Galois Representations 9
4.1 Column B: Number of Generators . . . 9
4.2 Column C: Generator Heights . . . 9
4.3 Column D: Regulator . . . 10
4.4 Column E: Galois Representation Surjective for All Primes? . . . 10

5 Table 3: Galois Exceptional Primes and Modular Data 11
5.1 Column B: Galois Representation Exceptional Primes . . . 11
5.2 Column C: Galois Image Type . . . 12
5.3 Column D: Index of $\Gamma_0(N)$ in $\mathrm{SL}_2(\mathbb{Z})$ . . . 13

6 Table 4: Modular Curves and L-functions **13**
6.1 Column B: Genus of $X_0(N)$ . . . 13
6.2 Column C: Number of Cusps . . . 14
6.3 Column D: Has Complex Multiplication? . . . 15
6.4 Column E: CM Discriminant . . . 15
6.5 Column F: Root Number . . . 16

7 Why Representation Theory Matters: A Summary **17**
7.1 The Langlands Program . . . 17
7.2 Local-Global Principle . . . 17
7.3 Congruences and p-adic Theory . . . 18
7.4 Modularity Lifting . . . 18

8 Conclusion **18**

1 Introduction: What Are We Looking At?

These four tables catalog various invariants of elliptic curves over $\mathbb{Q}$—smooth projective curves of genus 1 with a specified rational point. While geometrically simple objects, elliptic curves connect to deep areas of modern number theory: they admit a group structure (making them abelian varieties), they parametrize modular forms via the Modularity Theorem, and they carry rich Galois representations encoding arithmetic information.

The tables split this data into natural categories:

1. **Table 1**: Basic invariants—weight, torsion, rank
2. **Table 2**: Mordell-Weil generators and Galois representations
3. **Table 3**: Galois exceptional primes and modular groups
4. **Table 4**: Modular curves and L-functions

We also include data for Ramanujan's Δ function, a weight 12 modular form, which serves as a conceptual bridge between elliptic curves and more general modular forms.

2 Background: Elliptic Curves and Their Groups

2.1 Definition and Basic Structure

Definition 1. *An **elliptic curve** over $\mathbb{Q}$ is a smooth projective curve E of genus 1 with a specified rational point $O \in E(\mathbb{Q})$.*

The General Weierstrass Form

Every elliptic curve over $\mathbb{Q}$ can be written as a **general Weierstrass equation**:

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

where $a_1, a_2, a_3, a_4, a_6 \in \mathbb{Q}$. The five coefficients are traditionally indexed this way for historical reasons. The specified base point is $O = [0 : 1 : 0]$, the unique point at infinity on the projective closure. The smoothness condition (no singular points) imposes a non-vanishing condition on the *discriminant* Δ (defined below).

From these five coefficients one derives, by a fixed recipe, the **auxiliary b-invariants**:

$$\begin{aligned}
b_2 &= a_1^2 + 4a_2 \\
b_4 &= a_1a_3 + 2a_4 \\
b_6 &= a_3^2 + 4a_6 \\
b_8 &= a_1^2a_6 - a_1a_3a_4 + 4a_2a_6 + a_2a_3^2 - a_4^2
\end{aligned}$$

and then the **c-invariants**:

$$c_4 = b_2^2 - 24b_4, \qquad c_6 = -b_2^3 + 36b_2b_4 - 216b_6$$

and the **discriminant**:

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6.$$

The curve is smooth (non-singular) if and only if $\Delta \neq 0$.

Minimal Weierstrass Models

A given elliptic curve can be described by infinitely many Weierstrass equations, related by changes of variables. For arithmetic purposes it matters to choose coefficients that are as small as possible. A Weierstrass equation is called **minimal at a prime** p if the p-adic valuation $v_p(\Delta)$ is as small as possible among all Weierstrass equations for the same curve with integer coefficients.

Definition 2. *A **minimal (global) Weierstrass model** of E is a Weierstrass equation*

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in \mathbb{Z},$$

that is simultaneously minimal at every prime p. Such a model exists and is unique up to a restricted class of change of variables. The coefficient vector $[a_1, a_2, a_3, a_4, a_6]$ is the data stored in the Cremona database and reported in column B of Table 1.

Why minimal models matter: Different Weierstrass equations for the same curve give different values of Δ, but the minimal model minimizes $|\Delta|$ over all integral models. This makes computations—and comparisons between curves—well-defined and canonical.

Short Weierstrass Form

Over fields of characteristic different from 2 and 3 (and in particular over $\mathbb{Q}$), the terms a_1xy, a_3y, and a_2x^2 can be eliminated by a change of variables, yielding the **short Weierstrass form**:

$$E : y^2 = x^3 + Ax + B, \qquad A, B \in \mathbb{Q}.$$

This is more compact and convenient for many calculations. However, the short form is *not unique*: for any nonzero $u \in \mathbb{Q}$, the substitution $x \mapsto u^2x$, $y \mapsto u^3y$ gives a new short form $y^2 = x^3 + u^4A\,x + u^6B$ for the same curve. For this reason the tables use the canonical minimal general Weierstrass model rather than the short form.

Group Structure

The miracle: $E(\mathbb{Q})$ forms an **abelian group** under a geometric addition law. Given two points $P, Q \in E(\mathbb{Q})$, the line through P and Q intersects E in a third point R, and we define $P + Q$ to be the reflection of R across the x-axis. The point O serves as the identity.

We write $[n]P$ for the n-fold sum:

$$[n]P = \underbrace{P + P + \cdots + P}_{n \text{ times}}$$

2.2 The j-Invariant

The **j-invariant** is a single rational number attached to each elliptic curve that captures its isomorphism class. It is computed from the minimal Weierstrass model as follows. Using the b- and c-invariants defined above,

Definition 3. *The **j-invariant** of an elliptic curve E with minimal Weierstrass model $[a_1, a_2, a_3, a_4, a_6]$ is:*

$$j(E) = \frac{c_4^3}{\Delta}.$$

This is always a well-defined rational number (since $\Delta \neq 0$ for a smooth curve). Note that the formula uses the minimal model's coefficients—not the short form—so there is no ambiguity.

Key fact: Two elliptic curves over $\overline{\mathbb{Q}}$ are isomorphic if and only if they have the same j-invariant. Over $\mathbb{Q}$, the situation is more subtle—curves with the same j-invariant are related by twists.

Special values:

- $j = 0$: The curve has complex multiplication by $\mathbb{Z}[\omega]$ where $\omega = e^{2\pi i/3}$. In our tables, curve 27a1 has $j = 0$.
- $j = 1728$: The curve has complex multiplication by $\mathbb{Z}[i]$. In our tables, curve 32a1 has $j = 1728$.
- $j < 0$ in factored form (e.g., $j = -2^{12}$ for 11a1): Negative j-invariants occur for curves with multiplicative reduction at the relevant prime.

2.3 The Conductor

The **conductor** N is a positive integer measuring the arithmetic complexity of the curve. For each prime p, we reduce the Weierstrass equation modulo p:

Definition 4. *An elliptic curve E has:*

- ***Good reduction*** *at p if the reduced curve $\tilde{E}/\mathbb{F}_p$ is smooth (nonsingular)*
- ***Bad reduction*** *at p if $\tilde{E}/\mathbb{F}_p$ is singular (has a node or cusp)*

Bad reduction comes in two types:

- **Multiplicative reduction**: $\tilde{E}$ has a node (ordinary double point)
- **Additive reduction**: $\tilde{E}$ has a cusp

The conductor encodes this information:

$$N = \prod_{p \text{ bad}} p^{f_p}$$

where the exponent f_p is defined by:

$$f_p = \begin{cases} 0 & \text{good reduction} \\ 1 & \text{multiplicative reduction} \\ 2 + \delta_p & \text{additive reduction} \end{cases}$$

and $\delta_p \geq 0$ measures the "wildness" of the additive reduction (related to valuations of the minimal discriminant and j-invariant at p).

Interpretation: The conductor is both a geometric invariant (measuring singularities of reduction) and an arithmetic invariant (appearing in functional equations of L-functions). Smaller conductor indicates "better" arithmetic behavior.

2.4 The Cremona Label

Elliptic curves are organized into **isogeny classes**—equivalence classes under isogenies (surjective morphisms with finite kernel). The **Cremona label** has the form NXm:

- N = conductor
- X = letter (a, b, c, ...) indicating isogeny class
- m = number ordering curves in the class

Example: `11a1`, `11a2`, `11a3` are three curves of conductor 11 in the first isogeny class.

3 Table 1: Basic Arithmetic Invariants

3.1 Column A: Curve Label

As described above. The tables include all curves with $N \leq 54$ from the Cremona database, plus the special entry "Δ" for Ramanujan's cusp form.

3.2 Column B: Weight

This refers to the **weight of the associated modular form**. To understand this, we need the Modularity Theorem.

A **modular form** of weight k for a congruence subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function $f : \mathbb{H} \to \mathbb{C}$ on the upper half-plane satisfying:

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z)$$

for all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$, plus growth conditions at cusps.

A **cusp form** additionally vanishes at all cusps. The space of weight k cusp forms for $\Gamma_0(N)$ is denoted $S_k(\Gamma_0(N))$.

3.2.1 Newforms, Oldforms, and Levels

Before stating the Modularity Theorem, we must clarify crucial terminology about levels and newforms.

Definition 5. *For a modular form f, we say f* ***has level*** *N or* ***appears at level*** *N if $f \in S_k(\Gamma_0(N))$.*

A key observation: if $f \in S_k(\Gamma_0(N))$ and M is any multiple of N (i.e., $N|M$), then $f \in S_k(\Gamma_0(M))$ as well. This is because $\Gamma_0(M) \subseteq \Gamma_0(N)$, so the transformation property for $\Gamma_0(N)$ automatically implies the transformation property for the smaller group $\Gamma_0(M)$.

This means: **a single modular form appears at infinitely many levels**—all multiples of some minimal level.

Definition 6. *The space $S_k(\Gamma_0(N))$ decomposes as:*

$$S_k(\Gamma_0(N)) = S_k^{new}(\Gamma_0(N)) \oplus S_k^{old}(\Gamma_0(N))$$

where:

- $S_k^{new}(\Gamma_0(N))$ *is the space of* ***newforms*** *at level* N*—forms that do* not *come from any proper divisor of* N

- $S_k^{old}(\Gamma_0(N))$ *is the space of* ***oldforms*** *at level* N*—forms that already appeared at some level* M *with* $M|N$ *and* $M < N$

More precisely:

$$S_k^{\text{old}}(\Gamma_0(N)) = \sum_{\substack{d|N \\ d<N}} S_k^{\text{new}}(\Gamma_0(d))$$

A form is "new" at level N if it genuinely first appears at that level, not having come from a lower level.

Definition 7. *For a newform* $f \in S_k^{new}(\Gamma_0(N))$*, we call* N *the* ***conductor*** *of* f *(as a modular form). This is the minimal level at which* f *appears.*

Key distinction:

- A newform f **has conductor** N (unique, minimal, intrinsic)

- The same form f **appears at levels** $N, 2N, 3N, \ldots$ (infinitely many, as oldform for $M > N$)

Theorem 8 (Minimality of Newform Level)**.** *Let* $f \in S_k^{new}(\Gamma_0(N))$ *be a newform. Then:*

1. $f \in S_k(\Gamma_0(M))$ *if and only if* $N|M$

2. N *is the unique minimal positive integer such that* $f \in S_k(\Gamma_0(N))$

Why newforms matter: Newforms are Hecke eigenforms with well-defined arithmetic meaning. Oldforms are linear combinations of newforms from lower levels—they don't give new arithmetic information, just redundancy. To study the arithmetic of modular forms, we focus on newforms at their minimal level (conductor).

Example: Consider the newform f associated to the elliptic curve 11a1. This newform has conductor 11, meaning:

- $f \in S_2^{\text{new}}(\Gamma_0(11))$ (newform at level 11)

- $f \in S_2(\Gamma_0(22))$ (oldform at level 22, since $11|22$)

- $f \in S_2(\Gamma_0(33))$ (oldform at level 33, since $11|33$)

- $f \in S_2(\Gamma_0(11k))$ for any $k \geq 1$ (oldform when $k > 1$)

But we say f "has level 11" or "has conductor 11" because that's where it's new.

Theorem 9 (Modularity Theorem, Wiles-Taylor-Breuil-Conrad-Diamond)**.** *Every elliptic curve* $E/\mathbb{Q}$ *of conductor* N *corresponds to a weight 2 newform* $f_E \in S_2^{new}(\Gamma_0(N))$*.*

Unpacking the statement:

- The conductor of E (defined geometrically via bad reduction: $N = \prod_{p \text{ bad}} p^{f_p}$)

- **Equals** the conductor of f_E (defined as the minimal level where f_E appears as a newform)

- This equality is the content of the Modularity Theorem—two a priori different notions of "conductor" coincide

The correspondence: For primes $p \nmid N$, the p-th Fourier coefficient of $f_E = \sum_{n=1}^{\infty} a_n q^n$ (where $q = e^{2\pi i z}$) satisfies:

$$a_p = p + 1 - \#E(\mathbb{F}_p)$$

This connects the analytic object f_E to the arithmetic of E.

For elliptic curves, the weight is always $k = 2$. For Ramanujan's Δ function:

$$\Delta(z) = q \prod_{n=1}^{\infty} (1 - q^n)^{24} = \sum_{n=1}^{\infty} \tau(n) q^n$$

we have $k = 12$. The function Δ generates the 1-dimensional space $S_{12}(\mathrm{SL}_2(\mathbb{Z}))$.

Why weight matters: The weight determines transformation properties and growth conditions. Higher weight means faster decay at cusps. Weight 2 is special: it corresponds to differential forms on modular curves, linking to algebraic geometry.

3.3 Column C: Torsion Order

Definition 10. *A point $P \in E(\mathbb{Q})$ is a **torsion point** if it has finite order, i.e., there exists $n \geq 1$ such that $[n]P = O$.*

The set of all torsion points forms a subgroup $E(\mathbb{Q})_{\text{tors}} \subseteq E(\mathbb{Q})$.

The Mordell-Weil theorem states that $E(\mathbb{Q})$ is a finitely generated abelian group. By the structure theorem:

$$E(\mathbb{Q}) \cong \mathbb{Z}^r \oplus E(\mathbb{Q})_{\text{tors}}$$

where $r \geq 0$ is the **rank** (Column E) and $E(\mathbb{Q})_{\text{tors}}$ is the finite **torsion subgroup**.

Column C shows $|E(\mathbb{Q})_{\text{tors}}|$, the order of this torsion subgroup.

Theorem 11 (Mazur, 1977)**.** *For elliptic curves over $\mathbb{Q}$, the torsion subgroup is one of exactly 15 possibilities:*

$$\mathbb{Z}/m\mathbb{Z}, \quad m \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12\}$$

or

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2m\mathbb{Z}, \quad m \in \{1, 2, 3, 4\}$$

This is a remarkable rigidity theorem—over other number fields, many more structures are possible.

3.4 Column D: Torsion Structure

This gives the explicit group structure as a product of cyclic groups $\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$ where $n_i | n_{i+1}$ (invariant factors form).

Examples from the table:

- "$\mathbb{Z}/5\mathbb{Z}$": Cyclic group of order 5
- "$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$": Non-cyclic group of order 8
- "0" or "trivial": Trivial group (no torsion)

Why torsion matters: Torsion points are defined over $\mathbb{Q}$, so they're "easy" rational points. For cryptography, we want large rank (many generators) but small torsion. For Diophantine problems, torsion can obstruct or facilitate solutions.

3.5 Column E: Rank

The **rank** r is the $\mathbb{Z}$-rank of the free part of $E(\mathbb{Q})$. Geometrically, it's the number of independent rational points of infinite order.

Example 12. *The curve* `37a1` $: y^2 + y = x^3 - x$ *has rank 1. One generator is* $P = (0, 0)$*, and every rational point is of the form* $[n]P + T$ *where* T *is a torsion point and* $n \in \mathbb{Z}$*.*

The rank problem: Computing the rank is **hard**. The best algorithms (descent methods) work provably for small rank but can fail for larger rank. The Birch and Swinnerton-Dyer (BSD) conjecture predicts:

$$\text{rank} = \text{ord}_{s=1} L(E, s)$$

where $L(E, s)$ is the L-function (see Section 6.6). For rank 0 and 1, BSD is proven in many cases, making rank computation feasible.

In our dataset ($N \leq 54$), we have:

- Rank 0: 138 curves
- Rank 1: 3 curves (37a1, 43a1, 53a1)
- Rank ≥ 2: 0 curves

4 Table 2: Mordell-Weil Generators and Galois Representations

4.1 Column B: Number of Generators

For a rank r curve, we need exactly r points to generate the free part $\mathbb{Z}^r$ of $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}}$. Column B shows this count. For rank 0 curves, this is 0. For the three rank 1 curves, this is 1.

4.2 Column C: Generator Heights

Given a point $P = (x, y) \in E(\mathbb{Q})$ with $x = n/d$ in lowest terms, the **naive height** is:

$$h(P) = \log \max(|n|, |d|)$$

This measures arithmetic complexity but isn't well-behaved under the group law. The **canonical height** (or Néron-Tate height) is:

$$\hat{h}(P) = \lim_{n \to \infty} \frac{h([2^n]P)}{4^n}$$

The canonical height is a **positive definite quadratic form**:

- $\hat{h}(P) \geq 0$ with equality iff $P \in E(\mathbb{Q})_{\text{tors}}$
- $\hat{h}([m]P) = m^2 \hat{h}(P)$
- $\hat{h}(P + Q) + \hat{h}(P - Q) = 2\hat{h}(P) + 2\hat{h}(Q)$ (parallelogram law)

Quadratic form interpretation: The parallelogram law makes $\hat{h}$ a quadratic form. The associated symmetric bilinear pairing is:

$$\langle P, Q \rangle := \frac{1}{2}(\hat{h}(P + Q) - \hat{h}(P) - \hat{h}(Q))$$

For generators $P_1, \ldots, P_r$ of $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}}$, write any point as $P = \sum_{i=1}^{r} n_i P_i$ with $n_i \in \mathbb{Z}$. Then:

$$\hat{h}(P) = \sum_{i,j=1}^{r} n_i n_j \langle P_i, P_j \rangle$$

This is a homogeneous quadratic polynomial in $(n_1, \ldots, n_r)$ with coefficients $\langle P_i, P_j \rangle$. The matrix $H_{ij} = \langle P_i, P_j \rangle$ is positive definite.

Column C lists $\hat{h}(P_i)$ for each generator P_i. These are numerical approximations (typically accurate to 10^{-10} or better) computed using a finite approximation to the limiting process.

Why heights matter: Heights measure arithmetic complexity. Small height means "simple" rational point. Height theory connects to:

- Counting rational points of bounded height
- Effective versions of Mordell's theorem
- The regulator (next column)

4.3 Column D: Regulator

For generators $P_1, \ldots, P_r$ of $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}}$, the **regulator** is:

$$\text{Reg}(E/\mathbb{Q}) = \det\left(\langle P_i, P_j \rangle\right)_{1 \le i,j \le r}$$

For rank 1, the regulator equals the height of the generator: $\text{Reg} = \hat{h}(P_1)$. For rank 0, we set $\text{Reg} = 1$ by convention.

Why the regulator matters: It appears in the BSD conjecture:

$$\lim_{s \to 1} \frac{L(E,s)}{(s-1)^r} = \frac{\Omega_E \cdot \text{Reg}(E/\mathbb{Q}) \cdot \prod_p c_p \cdot |\text{Sha}(E/\mathbb{Q})|}{|E(\mathbb{Q})_{\text{tors}}|^2}$$

where Ω_E is a period, c_p are Tamagawa numbers, and Sha is the Tate-Shafarevich group. The regulator measures the "volume" of the lattice of rational points.

4.4 Column E: Galois Representation Surjective for All Primes?

Now we enter representation theory. For each prime ℓ, consider the ℓ-torsion:

$$E[\ell] = \{P \in E(\overline{\mathbb{Q}}) : [\ell]P = O\}$$

This is a 2-dimensional vector space over $\mathbb{F}_\ell$: $E[\ell] \cong \mathbb{F}_\ell^2$ as abelian groups.

The absolute Galois group $G_{\mathbb{Q}} = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on $E[\ell]$ by:

$$\sigma \cdot P = \sigma(x, y) \text{ for } P = (x, y) \in E[\ell]$$

where σ extends to coordinates. This action respects the group structure, giving a representation:

$$\rho_{E,\ell} : G_{\mathbb{Q}} \to \text{Aut}(E[\ell]) \cong \text{GL}_2(\mathbb{F}_\ell)$$

The question: Is $\rho_{E,\ell}$ surjective onto $\text{GL}_2(\mathbb{F}_\ell)$?

Column E answers: "Is $\rho_{E,\ell}$ surjective for *all* primes ℓ?"

- Y (Yes/True) = surjective for all ℓ
- N (No/False) = non-surjective for at least one ℓ

What does Galois action mean? The Galois group $G_{\mathbb{Q}}$ permutes the algebraic numbers. When it acts on $E[\ell]$, it permutes the ℓ-torsion points while preserving the group structure. Larger image of $\rho_{E,\ell}$ means Galois "mixes" the ℓ-torsion points more thoroughly.

How is "more" quantified? By the size $|\mathrm{im}(\rho_{E,\ell})|$ and structure of the image subgroup. Maximal mixing occurs when $\mathrm{im}(\rho_{E,\ell}) = \mathrm{GL}_2(\mathbb{F}_\ell)$, which has order:

$$|\mathrm{GL}_2(\mathbb{F}_\ell)| = \ell(\ell-1)^2(\ell+1)$$

Why this matters: Galois representations encode arithmetic information:

- **Local data**: For $\ell \nmid N$, the trace $\mathrm{tr}(\rho_{E,\ell}(\mathrm{Frob}_p))$ equals a_p, the coefficient in $f_E = \sum a_n q^n$
- **Torsion fields**: $\mathbb{Q}(E[\ell])$ is the fixed field of $\ker(\rho_{E,\ell})$
- **Modularity**: The representation connects to automorphic forms
- **L-functions**: Functional equations and special values

Definition 13. *An elliptic curve $E/\mathbb{Q}$ has **complex multiplication** (CM) if its endomorphism ring over $\overline{\mathbb{Q}}$ is strictly larger than $\mathbb{Z}$:*

$$End_{\overline{\mathbb{Q}}}(E) \supsetneq \mathbb{Z}$$

By theory, when this occurs, $End_{\overline{\mathbb{Q}}}(E) \cong \mathcal{O}_K$ is an order in an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$.

Theorem 14 (Serre)**.** *For non-CM curves over $\mathbb{Q}$, the representation $\rho_{E,\ell}$ is surjective for all but finitely many primes ℓ.*

Most entries in Column E are "N" because even one exceptional prime makes it false. Table 3 lists these exceptional primes.

5 Table 3: Galois Exceptional Primes and Modular Data

5.1 Column B: Galois Representation Exceptional Primes

This lists the finite set of primes ℓ where $\rho_{E,\ell}$ fails to be surjective. Notation: "2—3—5" means primes 2, 3, and 5.

Why non-surjectivity occurs:

To understand this, we need the subgroup structure of $\mathrm{GL}_2(\mathbb{F}_\ell)$.

Definition 15. *The **Borel subgroup** $B \subseteq GL_2(\mathbb{F}_\ell)$ is the subgroup of upper-triangular matrices:*

$$B = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : ad \neq 0 \right\}$$

Interpretation: B is the stabilizer of the line $\langle e_1 \rangle \subseteq \mathbb{F}_\ell^2$ (spanned by the first basis vector). Elements of B scale this line and act arbitrarily on the quotient space.

Definition 16. *A **Cartan subgroup** of $GL_2(\mathbb{F}_\ell)$ is a maximal abelian subgroup. There are two conjugacy classes:*

***Split Cartan**: The diagonal matrices*

$$C_s = \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} : ad \neq 0 \right\} \cong \mathbb{F}_\ell^* \times \mathbb{F}_\ell^*$$

This has order $(\ell - 1)^2$.

***Non-split Cartan**: The image of the embedding $\mathbb{F}_{\ell^2}^* \hookrightarrow GL_2(\mathbb{F}_\ell)$, where multiplication by elements of $\mathbb{F}_{\ell^2}$ acts as $\mathbb{F}_\ell$-linear maps on $\mathbb{F}_{\ell^2} \cong \mathbb{F}_\ell^2$. This has order $\ell^2 - 1$ and is cyclic.*

Explicit non-split Cartan: If $\mathbb{F}_{\ell^2} = \mathbb{F}_\ell[t]/(t^2 - d)$ where $d \in \mathbb{F}_\ell$ is not a square, then the element t acts as:

$$t \mapsto \begin{pmatrix} 0 & d \\ 1 & 0 \end{pmatrix}$$

and generates (with scalars) a cyclic subgroup of order $\ell^2 - 1$.

Now we can describe when $\rho_{E,\ell}$ is not surjective:

1. **Rational ℓ-torsion**: If $E(\mathbb{Q})[\ell] \neq 0$, then $\rho_{E,\ell}$ fixes a non-zero point $P \in E[\ell]$, so the image is contained in the stabilizer of $\langle P \rangle$, which is conjugate to a Borel subgroup.

   Example: Curves 11a1, 11a2, 11a3 all have 5-torsion over $\mathbb{Q}$ (visible in Column D of Table 1), so $\rho_{E,5}$ is not surjective.

2. **Rational ℓ-isogeny**: If there's an isogeny $E \to E'$ of degree ℓ defined over $\mathbb{Q}$, the image of $\rho_{E,\ell}$ is contained in the normalizer of a Cartan subgroup.

3. **Complex multiplication**: If E has CM, the extra endomorphisms force the image of $\rho_{E,\ell}$ to lie in the normalizer of a Cartan subgroup for all ℓ.

For small ℓ (especially $\ell = 2, 3$), there can be **exceptional subgroups**—subgroups that don't fit the Borel/Cartan pattern. For instance, $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ (order 6) has a richer subgroup lattice than larger cases. For $\ell \geq 5$ and non-CM curves, Serre's analysis shows the image is typically all of $\mathrm{GL}_2(\mathbb{F}_\ell)$, with exceptions coming from the cases above.

5.2 Column C: Galois Image Type

A classification of the Galois representation:

- "surjective": $\mathrm{im}(\rho_{E,\ell}) = \mathrm{GL}_2(\mathbb{F}_\ell)$ for all ℓ
- "non-surjective at [list]": Lists exceptional primes
- "CM": Curve has complex multiplication

CM curves are special:

- Over $\mathbb{Q}$, there are only 13 possible CM j-invariants, corresponding to imaginary quadratic fields of class number 1: discriminants $D \in \{-3, -4, -7, -8, -11, -12, -16, -19, -27, -28, -43, -67, -163\}$
- Galois representations are never surjective (image lies in normalizer of Cartan)
- L-functions have special values (related to Hecke characters)
- Heights are easier to compute

Examples in our table: Curves 27a1-27a4 have CM by $\mathbb{Z}[\omega]$ where $\omega = e^{2\pi i/3}$ (discriminant -3).

5.3 Column D: Index of $\Gamma_0(N)$ in $\mathrm{SL}_2(\mathbb{Z})$

The **congruence subgroup** $\Gamma_0(N) \subset \mathrm{SL}_2(\mathbb{Z})$ consists of matrices:

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{N} \right\}$$

The index is:

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = N \prod_{p|N} \left(1 + \frac{1}{p}\right)$$

Examples:

- $N = 11$ (prime): Index $= 11 \cdot \frac{12}{11} = 12$
- $N = 15 = 3 \cdot 5$: Index $= 15 \cdot \frac{4}{3} \cdot \frac{6}{5} = 24$
- $N = 1$ (Δ): Index $= 1$ (since $\Gamma_0(1) = \mathrm{SL}_2(\mathbb{Z})$)

Why this matters:

1. Modular forms perspective: The dimension of the space of weight k cusp forms grows like:

$$\dim S_k(\Gamma_0(N)) \sim \frac{k-1}{12}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$$

for large N. Larger index means larger space of modular forms.

2. Representation theory: The group $\Gamma_0(N)$ acts on the upper half-plane $\mathbb{H}$, and modular forms are functions on $\Gamma_0(N)\backslash\mathbb{H}$ with nice transformation properties. The index tells us:

- Size of the fundamental domain
- Number of coset representatives
- Complexity of Hecke operators

3. Modularity: The curve E corresponds to a newform $f \in S_2(\Gamma_0(N))$. This newform is an eigenfunction for Hecke operators, and its Fourier coefficients a_n encode arithmetic data about E. The level N and index control:

- Which Hecke algebras act
- Congruences between modular forms
- Galois representations attached to f

6 Table 4: Modular Curves and L-functions

6.1 Column B: Genus of $X_0(N)$

The modular curve $X_0(N)$ is defined as:

$$X_0(N) = \Gamma_0(N)\backslash(\mathbb{H} \cup \mathbb{P}^1(\mathbb{Q}))$$

the compactification of the quotient of the upper half-plane by $\Gamma_0(N)$.

As a Riemann surface, $X_0(N)$ has genus g computed via Riemann-Hurwitz:

$$g = 1 + \frac{[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]}{12} - \frac{\nu_2}{4} - \frac{\nu_3}{3} - \frac{\nu_\infty}{2}$$

where ν_2, ν_3 count elliptic points and ν_∞ counts cusps.

Why genus matters:

1. Dimension of modular forms: For weight 2:

$$\dim S_2(\Gamma_0(N)) = g \text{ for } N > 1$$

Example: $X_0(37)$ has genus 2, so there are 2 independent weight-2 newforms. These correspond to curves 37a1 and 37b1.

2. Rational points: The curve $X_0(N)$ itself is an algebraic curve over $\mathbb{Q}$. Its rational points $X_0(N)(\mathbb{Q})$ parametrize pairs (E, C) where:

- E is an elliptic curve over $\mathbb{Q}$
- $C \subset E$ is a cyclic subgroup of order N

By Mazur's theorem, for $N > 163$, we have $X_0(N)(\mathbb{Q}) = \{\text{cusps}\}$, so there are no non-cuspidal rational points.

3. Genus and complexity:

- Genus 0: Rational curve (e.g., $X_0(1), X_0(2), \ldots, X_0(10)$)
- Genus 1: Elliptic curve (e.g., $X_0(11), X_0(14), X_0(15)$)
- Genus ≥ 2: General type, Mordell conjecture (Faltings) applies

6.2 Column C: Number of Cusps

Cusps are points added in compactification, corresponding to $\Gamma_0(N)$-orbits in $\mathbb{P}^1(\mathbb{Q})$. The count is:

$$\#\{\text{cusps}\} = \sum_{d|N} \phi(\gcd(d, N/d))$$

where ϕ is Euler's totient function.

Examples:

- $N = 11$ (prime): 2 cusps
- $N = 30 = 2 \cdot 3 \cdot 5$: 12 cusps
- $N = 1$: 1 cusp (the point at ∞)

Why cusps matter:

1. Fourier expansions: Modular forms have Fourier expansions at each cusp. A cusp form $f \in S_k(\Gamma_0(N))$ vanishes at all cusps, which imposes strong constraints.

2. Rational points: Cusps are always rational points of $X_0(N)$, representing "degenerate" elliptic curves (nodal or cuspidal cubics).

3. Dimension formulas: The genus formula involves the number of cusps, connecting topology to arithmetic.

6.3 Column D: Has Complex Multiplication?

As defined in Section 4.5, a curve has CM if $\mathrm{End}_{\overline{\mathbb{Q}}}(E) \supsetneq \mathbb{Z}$.

Why modular form workers care about CM:

1. Special L-values: For CM curves, the L-function $L(E, s)$ factors as:

$$L(E, s) = L(\psi, s)$$

where ψ is a Hecke character of the imaginary quadratic field. This connects to:

- Class field theory
- Algebraic Hecke characters
- Explicit formulas for special values

2. Congruences: CM forms often satisfy congruences with non-CM forms modulo small primes, leading to:

- Connections between different L-functions
- Iwasawa theory
- p-adic L-functions

3. Explicit construction: CM curves can be constructed explicitly via the theory of complex multiplication:

- Given discriminant D, construct j-invariant using modular functions
- Reduces to class field theory of imaginary quadratic fields
- Enables explicit computation of Fourier coefficients

6.4 Column E: CM Discriminant

If E has CM, this is the discriminant $D < 0$ of the imaginary quadratic order $\mathrm{End}(E)$. Standard values: $D \in \{-3, -4, -7, -8, -11, -12, -16, \ldots\}$.

Examples from table:

- Curves 27a1-27a4: $D = -3$ (CM by $\mathbb{Z}[\omega]$, $\omega^3 = 1$)
- Curves 32a1-32a4: $D = -4$ (CM by $\mathbb{Z}[i]$, $i^2 = -1$)
- Curves 49a1-49a4: $D = -7$ (CM by $\mathbb{Z}[\frac{1+\sqrt{-7}}{2}]$)

Connection to class field theory: The field $K = \mathbb{Q}(\sqrt{D})$ has class number h_D. The 13 discriminants mentioned earlier ($D = -3, -4, -7, -8, -11, -12, -16, -19, -27, -28, -43, -67, -163$) are exactly those with $h_D = 1$ (class number one). For these discriminants, the CM elliptic curves have rational j-invariants.

6.5 Column F: Root Number

The L-function of E is:

$$L(E,s) = \prod_{p \nmid N} \frac{1}{1 - a_p p^{-s} + p^{1-2s}} \cdot \prod_{p|N} L_p(E, p^{-s})$$

where the second product accounts for bad primes.

This has an analytic continuation to all $s \in \mathbb{C}$ and satisfies a functional equation:

$$\Lambda(E,s) = w \cdot \Lambda(E, 2-s)$$

where $\Lambda(E,s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(E,s)$ is the completed L-function, and $w \in \{+1, -1\}$ is the **root number**.

Why the root number matters:

1. Birch and Swinnerton-Dyer conjecture: The BSD conjecture predicts:

$$w = (-1)^r$$

where r is the rank. Thus:

- $w = +1$ predicts even rank (including rank 0)
- $w = -1$ predicts odd rank (including rank 1)

In our table:

- 138 curves have $w = +1$ and rank 0 (consistent with BSD)
- 3 curves (37a1, 43a1, 53a1) have $w = -1$ and rank 1 (consistent with BSD)

2. Representation theory: The root number is a product of local signs:

$$w = \prod_{p \leq \infty} w_p$$

where each local factor $w_p = \pm 1$ encodes arithmetic data at p. For finite primes $p|N$:

- **Multiplicative reduction at** p: Typically $w_p = -1$ ("bad for parity")
- **Additive reduction at** p: Typically $w_p = +1$ ("good for parity")

What does "good/bad for parity" mean? Since $w = \prod_p w_p$ and we want to know the sign:

- Each $w_p = -1$ contributes a negative sign to the product
- If there's an *odd* number of primes with $w_p = -1$, then $w = -1$ (predicting odd rank)
- If there's an *even* number of primes with $w_p = -1$, then $w = +1$ (predicting even rank)

So "bad for parity" means $w_p = -1$ flips the sign, potentially changing even rank prediction to odd (or vice versa). "Good for parity" means $w_p = +1$ doesn't flip the sign.

3. Modular forms: The root number relates to the Atkin-Lehner involution W_N acting on $S_2(\Gamma_0(N))$:

$$W_N : f(z) \mapsto \frac{1}{\sqrt{N}} f(-1/(Nz))$$

The newform f_E is an eigenfunction: $W_N(f_E) = w \cdot f_E$.

4. Deep connections: Via the Modularity Theorem:

- Root number connects to Galois representations (via local Galois data)
- Relates to epsilon factors in Langlands program
- Appears in formulations of equivariant Tamagawa number conjectures

7 Why Representation Theory Matters: A Summary

Throughout these tables, representation-theoretic data appears repeatedly. Why do modular form workers care so much about Galois representations?

7.1 The Langlands Program

The overarching framework is the **Langlands program**, which proposes deep connections between:

- **Automorphic side**: Modular forms, automorphic representations
- **Galois side**: Galois representations, motives

For elliptic curves, the Modularity Theorem is a special case: the curve E (geometric object with Galois action) corresponds to the newform f_E (analytic object with Hecke action).

The Galois representation $\rho_{E,\ell}$ encodes how Galois acts on ℓ-torsion, while the modular form f_E encodes the same information through Fourier coefficients. The correspondence is:

$$\mathrm{tr}(\rho_{E,\ell}(\mathrm{Frob}_p)) \equiv a_p \pmod{\ell}$$

for primes $p \nmid \ell N$.

7.2 Local-Global Principle

Galois representations encode:

- **Global data**: Via $\rho_{E,\ell} : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_\ell)$
- **Local data**: Via restrictions $\rho_{E,\ell}|_{G_{\mathbb{Q}_p}}$ for each prime p

The local data determines:

- Reduction type at p (good, multiplicative, additive)
- Conductor exponent f_p at p
- Local root number w_p

Modular forms see this same data through Fourier coefficients and Hecke eigenvalues. The representation-theoretic formulation unifies these perspectives.

7.3 Congruences and p-adic Theory

When two modular forms have congruent Fourier coefficients modulo a prime p, their associated Galois representations are congruent modulo p. This leads to:

- **Iwasawa theory**: Growth of Selmer groups in $\mathbb{Z}_p$-extensions
- **p-adic L-functions**: Interpolating special values p-adically
- **Bloch-Kato conjectures**: Relating regulators to L-values via K-theory

Understanding the Galois representation modulo p is essential for these deep arithmetic applications.

7.4 Modularity Lifting

The proof of Fermat's Last Theorem (Wiles-Taylor) used **modularity lifting theorems**: given a Galois representation ρ that looks modular mod p (matches a modular form mod p), prove ρ is modular (comes from a modular form over characteristic 0). This requires:

- Detailed control over Galois cohomology $H^1(G_{\mathbb{Q}}, \mathrm{ad}^0(\rho))$
- Deformation theory of representations
- Understanding ramification and local conditions

The representation-theoretic data in Tables 2 and 3—exceptional primes, image structure, CM properties—is exactly the input needed for such theorems. To prove modularity of a Galois representation, one must understand:

- At which primes is it ramified? (Conductor)
- What is the local structure at ramified primes? (Reduction type)
- What are the irreducibility properties? (Image of ρ)
- Are there congruences with known modular forms? (Exceptional primes)

8 Conclusion

These four tables encode a rich tapestry of connections between:

- Algebraic geometry (elliptic curves as varieties)
- Group theory (rational points as abelian groups)
- Complex analysis (modular forms as holomorphic functions)
- Representation theory (Galois representations)
- Number theory (L-functions, BSD conjecture)

For researchers in modular forms, the representation-theoretic perspective is not optional—it's the language in which modern number theory is written. The Galois representations provide:

- A unified framework (Langlands program)
- Computational tools (modularity, congruences)
- Deep conjectures (BSD, Bloch-Kato)
- Connections to other areas (arithmetic geometry, algebraic topology)

Even for those primarily interested in modular forms as analytic objects, understanding the associated Galois representations is essential for grasping the deeper arithmetic meaning of Fourier coefficients, Hecke eigenvalues, and L-functions.

The data in these tables—conductor exponents, torsion structures, generator heights, exceptional primes, genus, root numbers—are not isolated curiosities but fundamental invariants that:

- Determine the structure of Galois representations
- Control dimensions of modular form spaces
- Predict ranks via the BSD conjecture
- Enable explicit computations of L-values
- Guide the search for rational points

Understanding what each column means, and why it matters, is the first step toward appreciating the profound unity underlying the arithmetic of elliptic curves.